

Relational Database Design from Basic Relationships to Research Funding

Adnan Khan

I completed four database labs and a research-funding assignment. The project covers relationship foundations, associative entities, a connected medical model, normalization and written business rules. I use corrected design summaries where the original notation was inconsistent.

Tools and environment

Entity-relationship diagrams; logical data modelling; primary/foreign and composite keys; associative entities; functional dependencies; first, second and third normal form.

Individual ITM500 coursework by Adnan Khan. March-April 2026. The evidence is logical design and analysis; no DBMS implementation is claimed.

Separate customers from their orders

I kept customer attributes in Customers and order-specific attributes in Orders. `customer_id` connects the order back to its customer, while `order_number` identifies the order. This avoids repeating the customer profile for every transaction.



Customers and orders. Source portfolio p. 6.

Apply consistent references in basic relationships

The early exercises include faculty/students, captains/ships and positions/players. The common design principle is to keep shared information with its entity and place the reference in the dependent record. For faculty/students, the corrected requirement is to use the same faculty identifier on both sides.



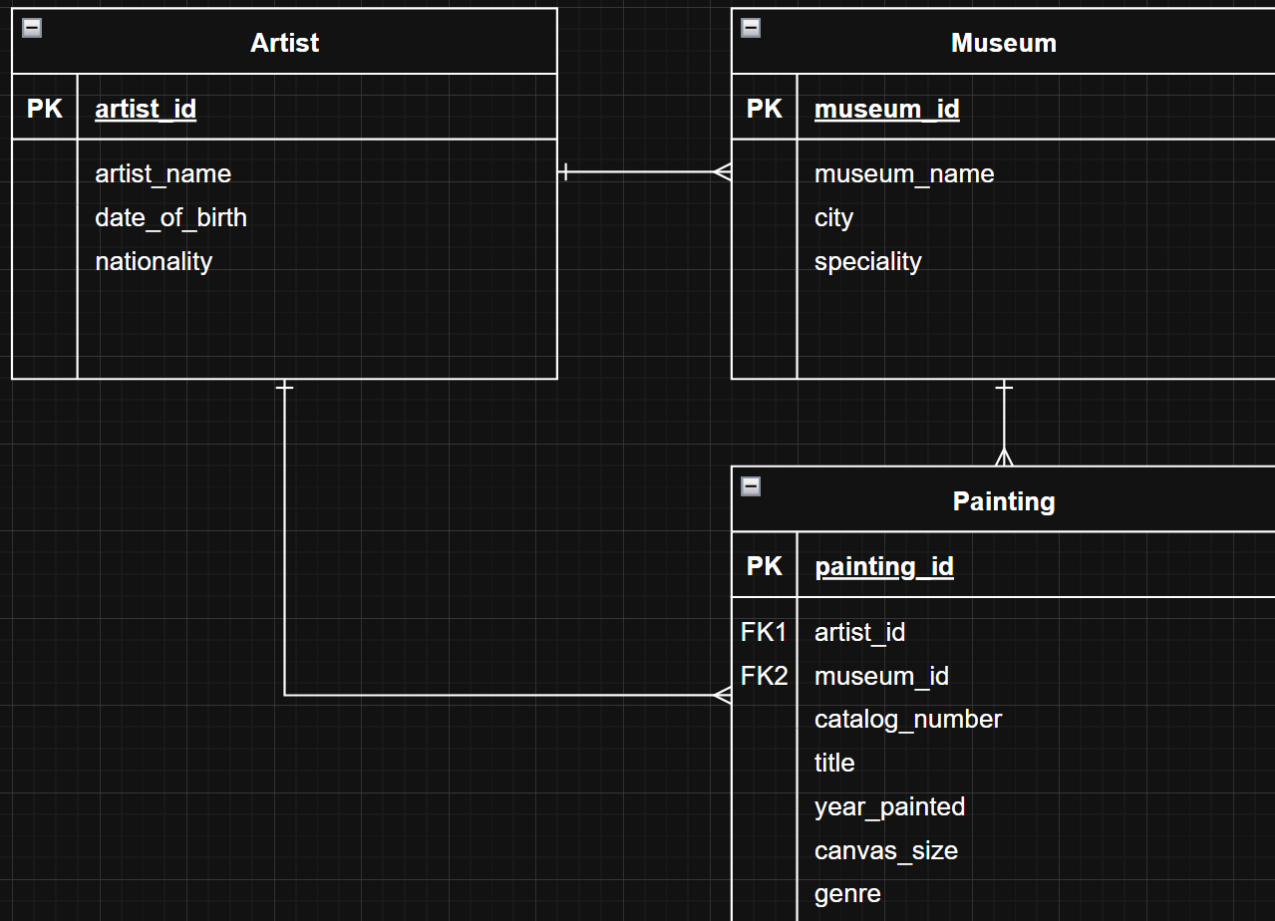
Captains and ships. Source portfolio p. 8.



Positions and players. Source portfolio p. 9.

Connect artists, museums and paintings

I modelled Painting as the record connecting an artist and a museum. Its foreign keys connect ownership/location context to the individual work. The design separates reusable artist and museum details from each painting's attributes.



Artists museums and paintings. Source portfolio p. 10.

Store facts about an association

The musician exercise places expertise level with the musician-instrument relationship. The library exercise places reservation date with the borrower-book relationship. A clear design needs a consistent relationship key, especially if the same pair can recur over time.

- Corrected conceptual structure: Musician → Expertise ← Instrument_Type; level belongs to Expertise.
- Reservation connects a cardholder and ISBN, with date as event information.
- Repeated reservations need an explicit event-identity rule before a physical key is chosen.

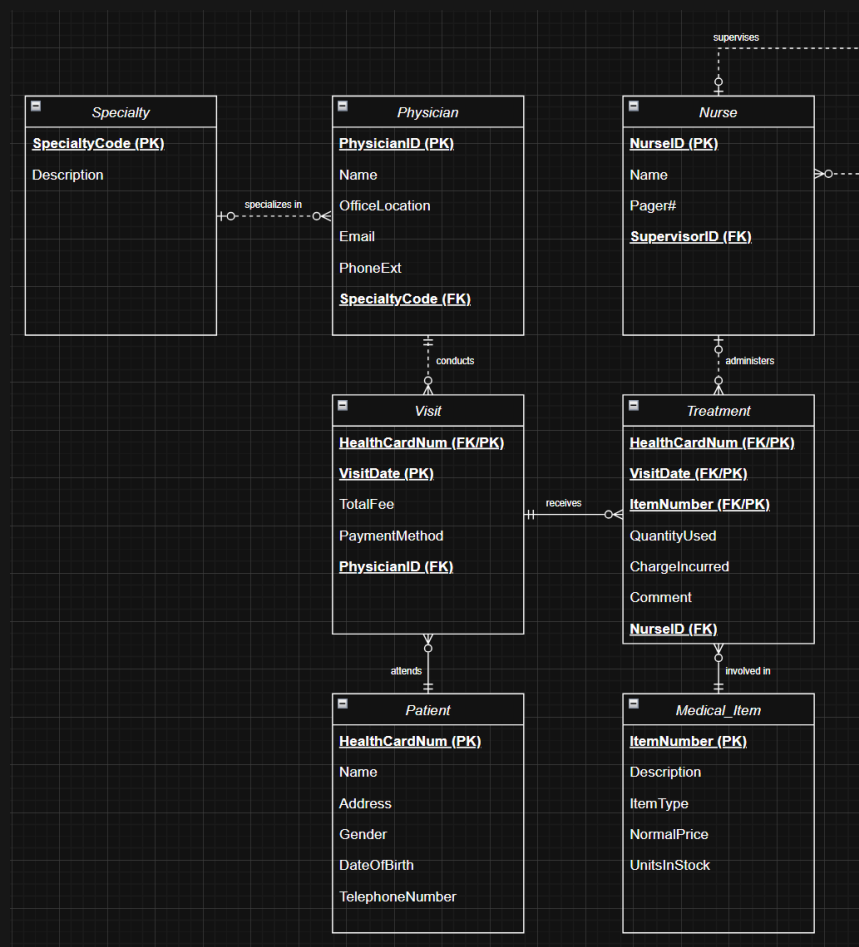
Model appointment and screening events

The patient appointment exercise connects a patient with a doctor; the screening exercise connects a movie with a theatre inside a complex. Date and time describe the event. Event identity and scheduling rules must be defined separately from the basic participant relationships.

- Allowing multiple visits on the same day requires a suitable encounter identifier.
- A screening key alone does not prevent overlapping bookings; that needs an explicit constraint or scheduling check.

Build the connected medical-centre model

I extended the relationships into a seven-entity medical model. A visit references the patient and physician; treatment entries reference the visit and a medical item. Quantity and charge describe the treatment entry, while item descriptions and prices stay in the catalogue.



Medical centre logical model. Source portfolio p. 17.

Follow the data behind a patient visit

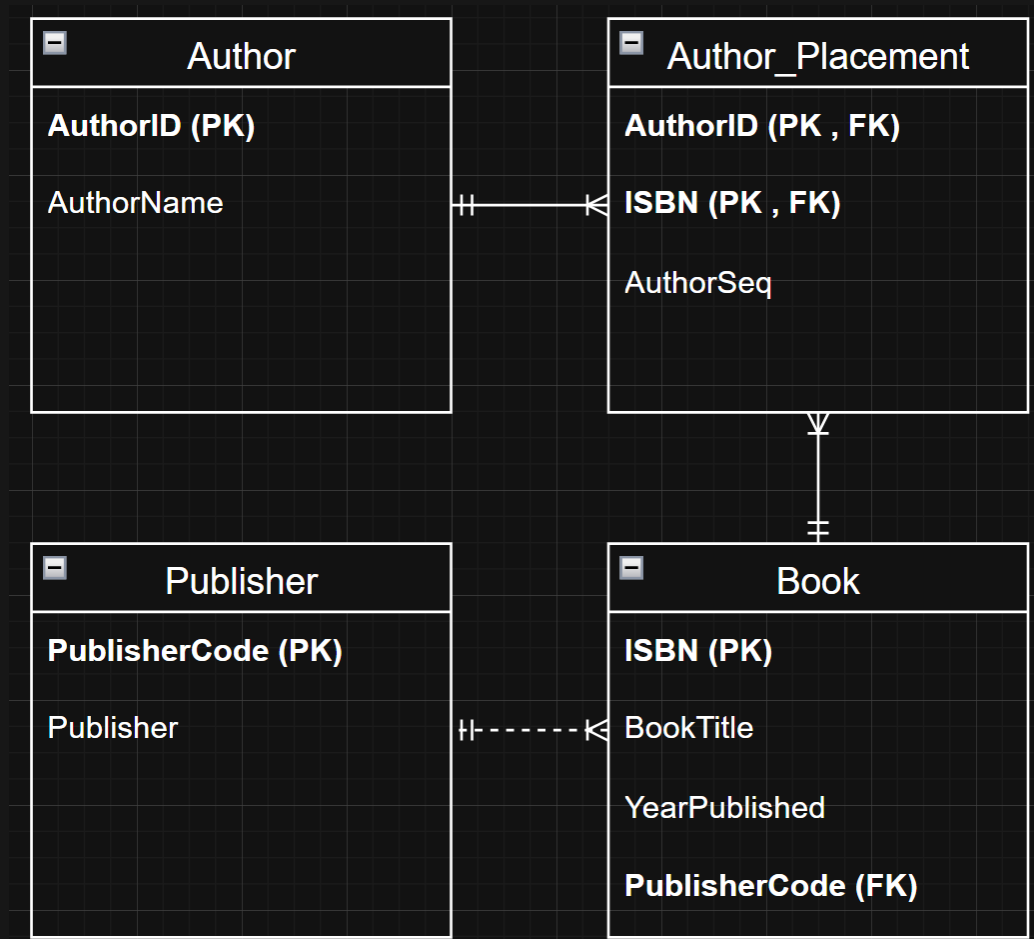
The patient record supplies identity details. Visit records capture the encounter and physician reference, then treatment rows describe what was provided. The model links physicians and departments so responsibilities can be traced without copying the same descriptions into each visit.

- The submitted model uses composite visit/treatment keys. A physical implementation would first settle how repeated encounters and repeated treatments are identified.

Normalize publication records

I decomposed publication data into Author, Publisher, Book and Author_Placement. Author_Placement connects AuthorID and ISBN and carries AuthorSeq. Publisher facts stay with Publisher, book facts with Book, and author facts with Author.

- 1NF makes the recorded values atomic.
- 2NF removes dependencies on only part of a composite key.
- 3NF separates facts that depend on another non-key attribute.



Normalized publication model. Source portfolio p. 21.

Define the research-funding scope

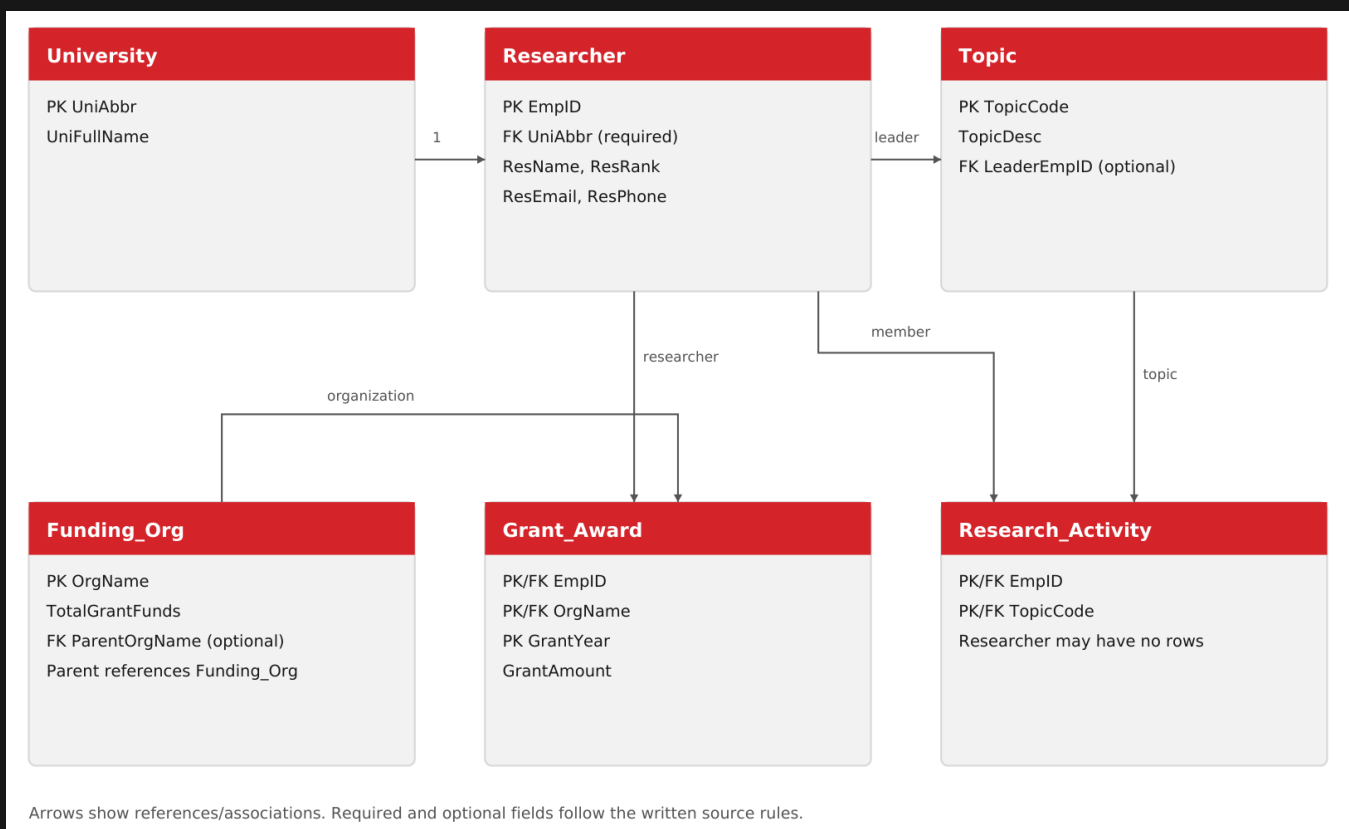
The assignment brings together universities, researchers, topics, funding organisations and grants. Researchers belong to a university; research participation connects researchers to topics. Funding organisations can sit in a parent hierarchy.

- The grant record carries researcher, organisation, year and amount.
- The topic-leader role is separate from general research participation.

Align keys and optionality with written rules

The corrected funding design identifies an annual grant by researcher, funding organisation and year. GrantYear is part of that key rather than a foreign key to an unspecified table. Topic leadership and the parent-organisation reference are optional where the written rules allow it.

- A researcher may exist without a Research_Activity row.
- Topic.LeadEmpID may be unassigned.
- Funding_Org.ParentOrgName may be unassigned and otherwise references Funding_Org.
- A repeated researcher/organisation pair is allowed in a different grant year.



Revised design summary based on the source portfolio; not an execution screenshot.

Plan constraint tests before implementation

I translated the design into proposed validation cases. These tests would check that the database enforces the written rules, not just that the diagram looks connected. They remain a test plan because the source contains no executed database tests.

- Reject an order with a missing customer reference.
- Reject a duplicate author-book association or duplicate annual grant key.
- Allow documented optional references and researchers with no topic rows.
- Test repeated events, conflicting screenings and hierarchy cycles against explicit rules.

What I learned

- Store an attribute with the record or relationship it actually describes.
- Composite keys need to match the real level of detail, including time where events recur.
- Check diagram notation against written rules before building a physical schema.

Source: the full ITM500 portfolio supplied for this project. Evidence is from the original coursework; this edition adds no new lab test results. Repeated captures and earlier drafts are consolidated. Revised design summaries are labelled, and sensitive identifiers are masked.